

Securing Biomedical Visual Data: An Efficient-net Deep Learning Framework for Copy-Move Tampering Detection

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Abstract

In the modern digital era, image authenticity has become very important, especially with the advancement in manipulation technologies such as copy-move forgery. This study proposes a powerful and efficient deep learning framework for detecting copy-move forgeries in large-scale and diverse image datasets. The suggested approach combines multiple essential elements: image prepossessing to reduce noise and standardize, data enhancement through conditional Generative Adversarial Networks (cGAN) for diversity increase in the dataset, and feature extraction through Histogram of Oriented Gradients (HoG) to obtain critical structural information. Recursive Feature Elimination (RFE) is used to optimize the feature set further, enhancing the model's attention on impactful features. EfficientNet, a computationally efficient convolutional neural network, serves as the core detection model on which it learns to distinguish between forged and original images. The approach is exhaustively tested on three benchmark datasets, CASIA1, MICC-F220, and CoMoFoD, with stunning performances reporting detection accuracies of 98.25%, 99.8%, and 99.01% respectively. The suggested method surpasses several current models and creates a very reliable and scalable solution for digital image forgery detection in practical applications.

Keywords: EfficientNet, Large Image Datasets, Recursive Feature Elimination, Copy-move forgery detection, Generative Adversarial Networks

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1. Introduction

In the digital era, the authenticity and integrity of images have been a matter of grave concern, giving rise to increased importance of digital image forensics [1]. Some usual techniques for creating fake images are coloring, retouching, copy-move forgery, digital watermarking, splicing, and resampling [2-9]. Of these techniques, the most common type of image forgery is Copy Move Forgery (CMF), where a part of an image is copied and reintroduced into the same image to modify visual information [10,11]. This form of forgery is challenging to identify due to the replicated areas possessing hues, textures, and statistical features similar to those of the original picture [12]. Ensuring the reliability of digital images is crucial in fields such as newspapers, digital forensics, scientific research, and medicine, where manipulated

content can mislead decision-making processes [13]. Figure 1 below consists of two sets of images, showing the original images and their respective forged version, demonstrating CMF in images.

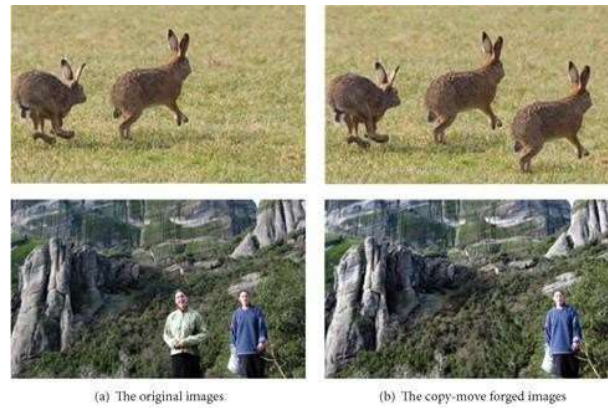


Figure 1. Illustration of CMF in images: (a) Original images, (b) Forged images [14].

Detecting copy-move forgeries is a complex problem due to several factors. Variations in illumination, texture consistency, geometric transformations (such as rotation, scaling, and affine transformations), and computational complexity make forgery detection difficult, especially in large datasets [15,16]. Traditional methods utilize block-based and keypoint-based extraction of features to detect duplicated areas [17]. Block-based techniques, like Discrete Cosine Transform (DCT) [18] and Discrete Wavelet Transform (DWT) [19], segment the image into uniform-sized blocks and extract statistical or frequency domain features to detect similarities [20]. However, these methods are sensitive to noise and geometric transformations.

On the other hand, key point-based approaches, including Scale-Invariant Feature Transform (SIFT) [21] and Speeded-Up Robust Features (SURF) [22], detect distinctive key points in an image and match them to identify duplicated regions [23]. While key point-based techniques are more robust to transformations, they often struggle with detecting small or texture less forgeries. Additionally, both approaches face limitations in processing high-resolution images efficiently, making them less practical for large-scale datasets.

Deep learning-based approaches have emerged as powerful alternatives to traditional methods, offering superior feature extraction and classification capabilities [24]. Unlike handcrafted feature-based techniques, deep learning models automatically learn hierarchical representations of images, making them more effective at identifying subtle and complex forgery patterns [25,26]. Convolutional Neural Networks (CNNs), in particular, have demonstrated significant achievement in image forensics, but their performance depends heavily on the quality of feature extraction and network architecture [27]. EfficientNet is a CNN architecture known for optimizing feature extraction with high computational efficiency. Balancing network constraints improves Accuracy while reducing parameters [28]. Efficient Net's ability to capture fine-grained image details and robust features enhances detection performance, making it a promising solution for CMF detection.

The constantly increasing CMF is already hard to detect due to its diversity in forgery types, large datasets, and geometry changes. Existing approaches are inefficient because they rely on traditional methods that are static and weak, leading to high costs and a failure to meet expectations. High-level requirements remain unmet for working with diverse datasets while spending fewer computing resources effectively. This research aims to build a robust and efficient system for image forgery detection and localization using deep learning techniques that utilize an

Efficient Net backbone with image preprocessing, augmentation, extraction, data feature, and copy-move image forgery detection in complex scenarios. The primary objectives of this paper are:

- Developing a robust Forgery Detection Model using the EfficientNet architecture for accurately detecting copy-move forgeries in images from diverse datasets.
- Enhancing model generalization through data augmentation using cGANs to simulate various forgery types.
- Improving the model's efficiency by incorporating advanced feature extraction techniques and feature selection methods.
- Evaluating the proposed forgery detection framework on diverse real-world image datasets using various metrics to test robustness and generalizability.

The other parts of this study are as follows: Section 2 provides a literature review. Section 3 outlines the dataset employed and the methodology of the study.

2. Literature review

Due to the many fabricated images, identifying forgery in digital media is essential today. **Alfraihi et al. [29]** devised a new method combining fusion-based transfer learning with Tasmanian Devil Optimizer (TDO) to improve video CMF detection. In this study, the optimized Elman recurrent neural network was used, which was trained to classify the forged video frames and recorded a classification accuracy of 95.26 % for the GRIP dataset and 92.67% for the VTD dataset. In addition, **Ustubioglu et al. [30]** focused on audio CMF detection that used a CNN model to classify graph images from audio filter spectrograms. The model performed exceptionally well across different datasets, achieving an optimal accuracy of 97% on the D4 dataset. Moreover, **Arivazhagan et al. [31]** employed transfer learning and a customized CNN model to identify copy-move forgeries and achieved a True Positive Rate of 100% on various forgery-related datasets.

Enhanced image forgery detection was demonstrated by **Diwan et al. [32]**, who introduced a neural network-driven Multiscale Detector that outperformed in key-point detection and descriptor extraction. The method showed a remarkable F1-score of 96.87% on the MICC-220 dataset, demonstrating robustness under various post-processing manipulations. A similar study was presented by **Kaushik et al. [33]**, who employed CNN-based deep features and applied Neighborhood Component Analysis to the feature extraction process, ultimately classifying with a Support Vector Machine (SVM) classifier for detection of copy-move forgery at 98.0% accuracy on the MICC-F220 and 97.62% on CASIA 1. After this, **Alhaidery et al. [34]** suggested a multi-stage approach that applies SURF-HOG detectors, Slic segmentation, and overlapping Zernike moments to enhance CMF detection. It achieved a desirable positive rate of 93.75% and was significantly robust against photometric attacks and geometric transformations.

To further address computational complexity and Accuracy, **Fu et al. [35]** developed a method that combined SURF and density-based clustering for enhanced forgery localization. The study achieved 95.23% accuracy using the CoMoFoD dataset with significantly reduced computational complexity. Further, **Patel et al. [36]** introduced a hybrid Long Short-Term Memory (LSTM)-CNN network and SVM classification for the detection of CMF with an accuracy of 94.7% on the CASIA database with patch rotation included. Moreover, a study by **Rodriguez-**

Ortega et al. [37] addressed the increasing problem of high-quality fake photos on social media. The study emphasized the importance of recognizing models based on deep learning to prevent digital forgeries successfully. Further, to identify splicing and copy-move forgeries, **Islam et al. [38]** laid out a dual-forgery detection strategy. The proposed approach achieved an average detection rate of more than 90% on both CASIA datasets.

2.1 Research gap

- Several classical techniques lack computational efficiency and effective feature selection and extraction techniques [32].
- Previous methods struggle to handle forgery features across multiple scales, particularly in cases where forgeries might involve low-resolution or subtle changes that are hard to detect [29][35].
- Many existing methods show degraded performance when applied to large-scale image datasets [34].
- Most current approaches rely on limited datasets, which lack the correct representation of the real world with diverse images and complex forgery techniques [30].

3. Methodology

This section explains the methodology used in this study, including the description of the data set, techniques used, proposed methodology, and evaluation measures.

3.1 Dataset description

This subsection thoroughly introduces the datasets employed to detect image forgery. Examples of sample images for each of the datasets are provided in Figure 2 below, and each of the datasets is described as follows:

- **CASIA Dataset**

It is a significant repository for image fraud detection, comprising over 7 thousand images with diverse forgery types. It provides diverse real-world pictures for training and evaluating models, making it appropriate for identifying various image alterations [39].

- **MICC-F220 Dataset**

It comprises 220 images, including both authentic and counterfeit images. The fabricated pictures are produced using copy-move modification, encompassing rotating, translating, scaling, or any combination. The post-processing methods are utilized to cover up evidence of counterfeiting. Certain pictures display an isolated occurrence of CMF, while others exhibit several instances [40].

- **CoMoFoD Dataset**

It comprises authentic and counterfeit pictures, each with its corresponding ground truth. The fabricated pictures are produced via five alteration techniques: rotation, translation, scaling, conjunction, and deformation, with 40 photos allocated for each approach. Furthermore, post-processing features such as compression, noise introduction, brightness modification, colour diminution, contrast amplification, and smoothing are implemented in all generated pictures. The collection comprises almost 4,000 counterfeit images for examination [41].



Figure 2. Sample images from CASIA, MICC-F220, and CoMoFoD datasets, respectively.

3.2 Techniques used

The following methods are used in the proposed approach:

3.2.1 HoG

HoG is a feature extraction technique that captures the distribution of edge directions in an image, which helps detect local patterns such as edges and textures. HoG computes local gradient magnitudes and orientations at each pixel and then forms histograms of gradient directions within local cells [42]. The resulting feature vector is invariant to geometrical alterations. The formula for HoG involves computing the gradient at each pixel using:

$$G(i, j) = \sqrt{G_x(i, j)^2 + G_y(i, j)^2} \quad (1)$$

$$\theta(i, j) = \tan^{-1} \left(\frac{G_y(i, j)}{G_x(i, j)} \right) \quad (2)$$

Where G_x and G_y represent the gradients of the image in the x and y directions, i and j denote the row and column indices, while θ designates the angle of the gradients [43]. These partitions allow the creation of orientation histograms, which describe the edges and textures of the image. In this research, the images' relevant information is captured using a HoG feature descriptor to detect possible regions of interest, such as areas subjected to copy-move modifications.

3.2.2 Recursive Feature Elimination (RFE)

RFE is a model-tuning strategy for feature selection that strives to increase model accuracy while minimizing overfitting by removing the least important features, one at a time, for every iteration [44]. The approach initializes with all features available, builds a model, and analyzes the features concerning their importance before discarding the least essential one. This procedure is repeated until the desired number of characteristics is achieved [45]. A feature importance rank usually accompanies a model-specific equation, such as weight coefficients for linear models or feature importances on decision trees. For instance, in a linear model, the importance of a feature f_i could be determined by its coefficient.

$$w_i = \text{coefficient of } f_i \quad (3)$$

where w_i is the weight associated with the feature f_i .

RFE is employed in this study to select the most significant HoG features, thus focusing on the most crucial elements for identifying forged areas in images.

3.2.3 cGAN

It is used for data augmentation by generating new samples conditioned on specific input data. This method employs a network of generators to produce synthetic pictures and a discriminator network to differentiate between actual and synthetic pictures [46]. The generator attempts to confuse the discriminator, whereas the discriminator attempts to identify photos as authentic or counterfeit correctly. A loss function for cGAN can be expressed as [47]:

$$L(G, D) = \mathbb{E}_{y \sim p(y)} [\log(D(y))] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))] \quad (4)$$

where G is the generator, D is the discriminator, y are real images and labels, and z is the latent input to the generator. This adversarial training results in the generator learning to produce highly realistic variations of the input data, valid for training deep models.

In this study, cGAN is used to augment the training dataset by generating new forged images, which helps the model generalize better to various manipulation scenarios, ultimately improving its performance in detecting forgery.

3.2.4 EfficientNet

EfficientNet is a type of CNN architecture optimized for performance and efficiency [48]. The model scales uniformly in resolution width and depth, allowing it to perform better while upholding computational efficiency. The EfficientNet architecture is structured from depth, width, and resolution scaling [49]. The standard cross-entropy loss function is given as:

$$L = - \sum_{i=1}^C p_i \log(\hat{p}_i) \quad (5)$$

Where, p_i represents the real label for class i, $\hat{p}_i$ is the anticipated likelihood for class i, and C signifies the overall number of categories.

EfficientNet surpasses the performance of many other CNN architectures by using fewer parameters without compromising the classification accuracy achieved [50]. In this study, EfficientNet is customized to categorize images as forged or non-forged. Using its efficiency and Accuracy, EfficientNet might significantly aid in effective copy-move forgery detection in images.

3.3 Proposed methodology

The forgery detection process in images starts with data gathering, where images are obtained from publicly available datasets that include both forged and non-forged samples. This provides diversity and a complete training set. Data preprocessing is then done, which includes image resizing, normalization for uniform representation, noise reduction to improve clarity, and image conversion to Grayscale if colour information is not required for the detection process. For better performance of the model, data augmentation is also employed through conditional cGANs to create variations of the images available by inducing different changes, flipping, cropping, etc., to enable the model to generalize more across various forms of forgeries. Feature extraction methods like HoG are also utilized to extract primary structural features like edges and texture that are fundamental in identifying the forged regions, as presented in Figure 3.

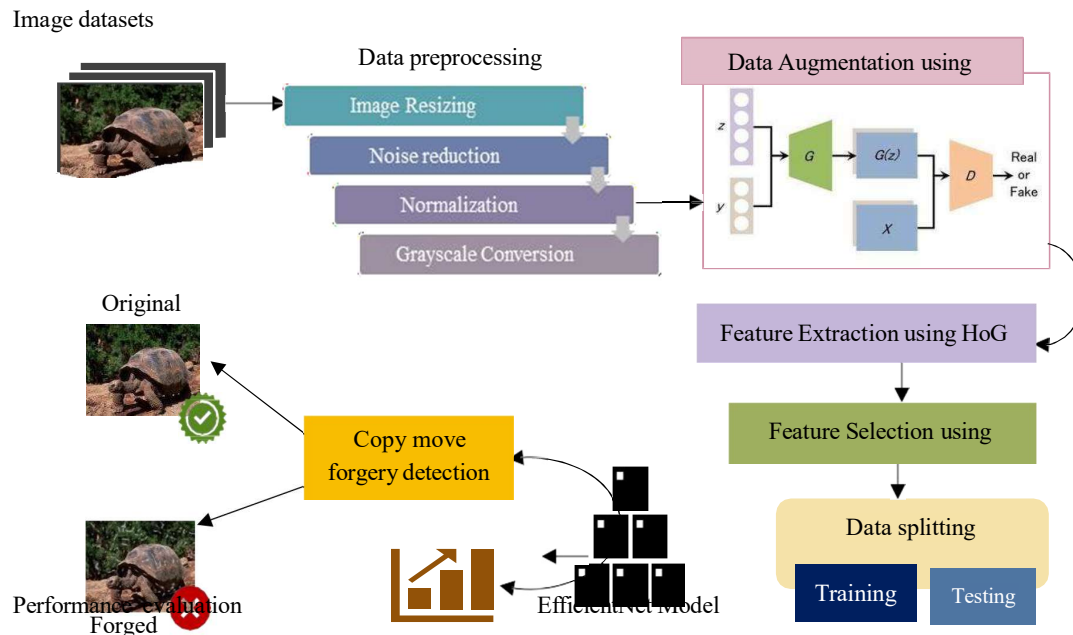


Figure 3. Proposed architecture.

To ensure optimal detection efficiency, model computation is controlled using RFE-based feature selection. The dataset is split into train and test sets while guaranteeing a balanced distribution of forged and non-forged images to evaluate the model without bias. The model training stage employs a deep learning architecture, EfficientNet, which is used for image forgery detection to classify images into original and forged. Efficient Net is fine-tuned with an appropriate loss function like the binary cross-entropy for enhanced learning. A loss-refined model is assessed against a variety of important performance metrics. Along with these, their computational efficiency is also checked regarding inference speed and resource consumption. Lastly, the model is continuously updated against new forgery techniques while managing the loss of Accuracy over time.

3.4 Proposed model architecture

Figure 4 shows the architecture of the Enhanced EfficientNet model tailored for image forgery identification. It begins with an Input Image of size (224x224x3) and goes through a Stem Layer (Conv2D) for low-level feature processing. Then, the network's backbone consists of several MBConv blocks (Mobile Inverted Bottleneck Convolutions), which are intended to achieve high efficiency in capturing the spatial hierarchies and patterns in the image. To achieve the optimum feature extraction with the least computation possible, these blocks use depthwise separable convolution, Squeeze-and-Excitation (SE) blocks, and Swish as the activation function. With each block the image goes through, the network captures higher and higher-level features for the forgery detection task.

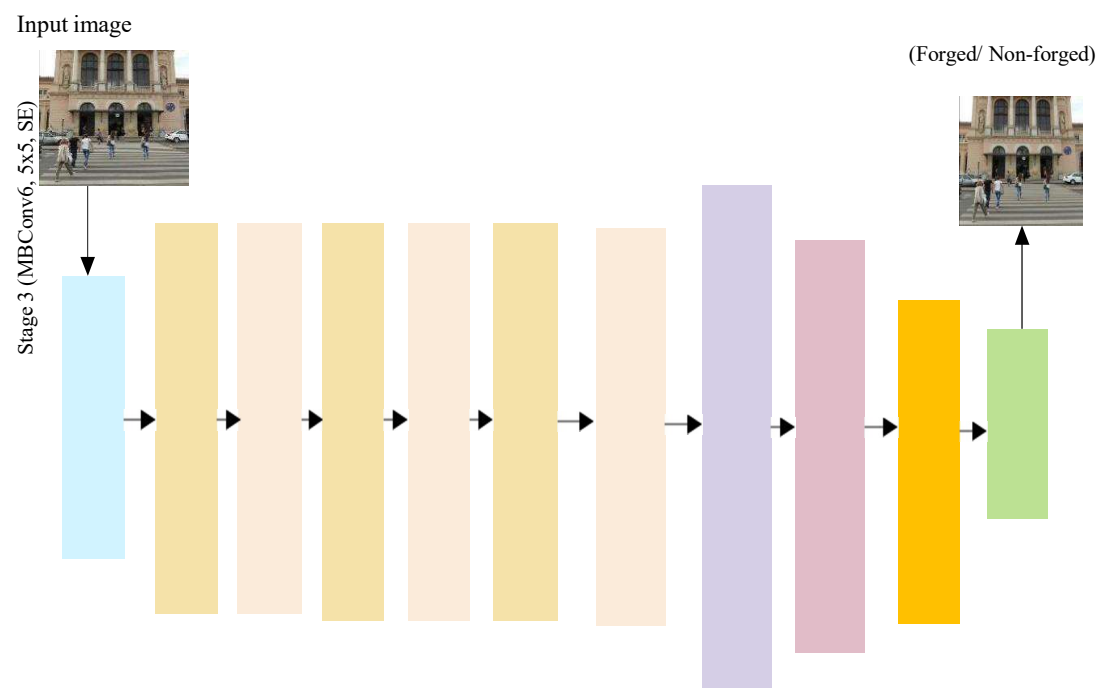


Figure 4. The architecture of the proposed efficient net model.

After the convolutional layers, the extracted features undergo refinement through a Final Conv Layer (1x1, BatchNorm, Swish), which improves the feature maps. After the Global Average Pooling (GAP) layer reduces the spatial dimensions of the feature maps into a vector representation, it transforms them into a set of features. This vector is passed into a Fully Connected Layer that links the learned features to the assigned probabilities for classification. The Output Layer analyzes the image to assess whether it is forged or not, thus resulting in binary classification. With efficiency at the core, this multi-faceted methodology enhances forgery detection without compromising performance.

3.5 Proposed algorithm

Algorithm: Efficient-net-based Copy Move Forgery Detection approach

Start

Step 1: Data Collection

Let D be the dataset consisting of three subsets:

$= \{ 1, 2, 3 \}$
where: D1 = CASIA dataset, D2 = MICC-F220 dataset, D3 = CoMoFoD dataset
The dataset consists of images Labelled as forged F or non-forged N:
 $= (,)$, $\{ , \}$

Step 2: Data Preprocessing

- Image Resizing: Transform each image To a fixed size (H, W) (e.g., 224×224):
 $' = \square\square(,)$

- Normalization: Scale pixel values to [0,1] using:

$$I' = \frac{I - \min(I)}{\max(I) - \min(I)}$$

- Noise Reduction: Apply Gaussian filter G with standard deviation σ :

$$I'' = (I * G)$$

- Grayscale Conversion (if required):

$$I''' = 0.2989 I_R + 0.5870 I_G + 0.1140 I_B$$

Step 3: Data Augmentation using cGAN

Let G and D be the generator and discriminator of cGAN, respectively. The augmented image I_a is generated by:

$$I_a = (I, z)$$

Where z is a random noise vector.

The augmentation transformations are:

- Rotation: $I_a = \text{rot}(I, \theta)$ ($\theta \in (-30^\circ, 30^\circ)$)
- Flipping: $I_a = \text{flip}(I)$ ($\text{flip} \in \{\text{horizontal}, \text{vertical}\}$)
- Scaling: $I_a = \text{scale}(I)$ ($\text{scale} \in (0.8, 1.2)$)
- Cropping: Extract a random patch P of size (h,w), where h and w are randomly chosen
- Brightness & Contrast: $I_a = \alpha I + \beta$, where α and β are random scalars

Step 4: Feature Extraction using HoG

For each image I , compute the gradient magnitudes $M(x,y)$ and orientations $\Theta(x,y)$:

$$M(x,y) = \sqrt{(I_x)^2 + (I_y)^2}$$

$$\Theta(x,y) = \arctan\left(\frac{I_y}{I_x}\right)$$

The HoG descriptor H is computed over a grid of cells:

$$H = \sum_{i,j} M(i,j) \cdot \sin(\Theta(i,j))$$

Step 5: Feature Selection using RFE

Define feature set $F = \{1, 2, \dots, n\}$. RFE recursively eliminates the least important feature based on model weight ranking:

$$\begin{aligned} \text{Importance} &= \max \sum (W(f)) \\ & \end{aligned}$$

where $W(f)$ represents feature importance.

Step 6: Dataset Splitting

Split dataset D into training () and testing ():
 , = (, 0.8)
Ensure balance:

 () $\approx$ ()

Step 7: Model Training and Forgery Detection using EfficientNet
Define the EfficientNet model M parameterized by θ :

' = (;)
Loss Function: Use Binary Cross-Entropy (BCE):

1
= - $\sum [y' \log(y) + (1 - y') \log(1 - y)]$
=1

Step 8: Performance Evaluation

Evaluate the model's performance in terms of correctly identifying forged and non-forged images and computational efficiency.

Step 9: Continuous Monitoring and Model Updating

Regularly monitor model performance over time to catch any degradation in Accuracy arising from the arrival of new, unseen data or changes to the very types of images being used.

End

4. Result and Discussion

This section describes the results and discussion based on the suggested EfficientNet-based framework for copy-move forgery detection. The rigid preprocessing of the model and highly efficient feature extraction with HoG, coupled with an ideal selection of attributes with RFE, showed consistent validity across the CASIA1, MICC-F220, and CoMoFoD datasets. Detection performance was further boosted when the framework was tested using cGAN-augmented data, proving the framework's reliability and versatility in adapting to different image sets while detecting the forged area.

4.1 Evaluation metrics

In this section, describe the evaluation metrics employed in the proposed study.

- **Accuracy:** To evaluate the Accuracy of the proposed model in detecting forged images.

$$= \frac{+}{+ + +} \tag{6}$$

where

TP is the number of True Positives, TN is the number of True Negatives, FP is the number of False Positives, and FN is the number of False Negatives.

- **Precision, Recall, and F1-Score:** These metrics must be calculated to study the model correctly classifying forged and non-forged images.

$$= \frac{+}{+ + +} \tag{7}$$

$$= \frac{+}{+ + +} \tag{8}$$

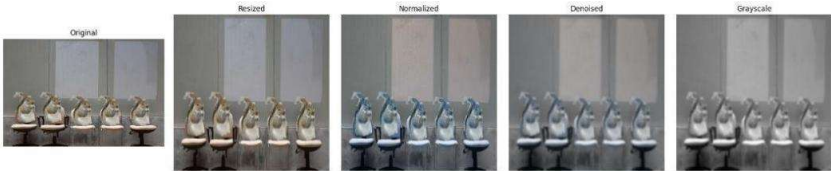
$$l = 2 \times \square \square \square \times \square \tag{9}$$

- **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):** The model's capacity to discriminate between authentic and forged images is evaluated by calculating the AUC-ROC curve.

- **Computational Efficiency:** The analysis of computational efficiency can be done by considering the time consumed during and in the model testing phase and the approach the model takes to handle heavy computation in a resource-economical way.

4.2 Data preprocessing

The main preprocessing steps of the CASIA, CoMoFoD, and MICC-F220 datasets are listed in Figure 5 (a-c), respectively. All sequences begin with the Original Image, which is the original and unprocessed input and, not surprisingly, possesses variation concerning resolution, illumination, and noise. First, the Resize transformation diminishes each image to a uniform size (224×224) to ensure uniformity within the dataset. Then, the Normalization operation adjusts pixel intensities to a uniform range, supporting numerical stability and facilitating quicker learning on the model's part. Gaussian-based Denoising is applied as a preliminary step towards eliminating visual noise and achieving better clarity, smoothing the image while preserving salient structural information. Finally, the images are, optionally, converted to Grayscale, diminishing the data and computational complexity and emphasizing intensity patterns, eliminating color. These preprocessing steps refine the quality of the visual data and establish a strong foundation for reliable feature extraction and subsequent forgery detection.



(a)

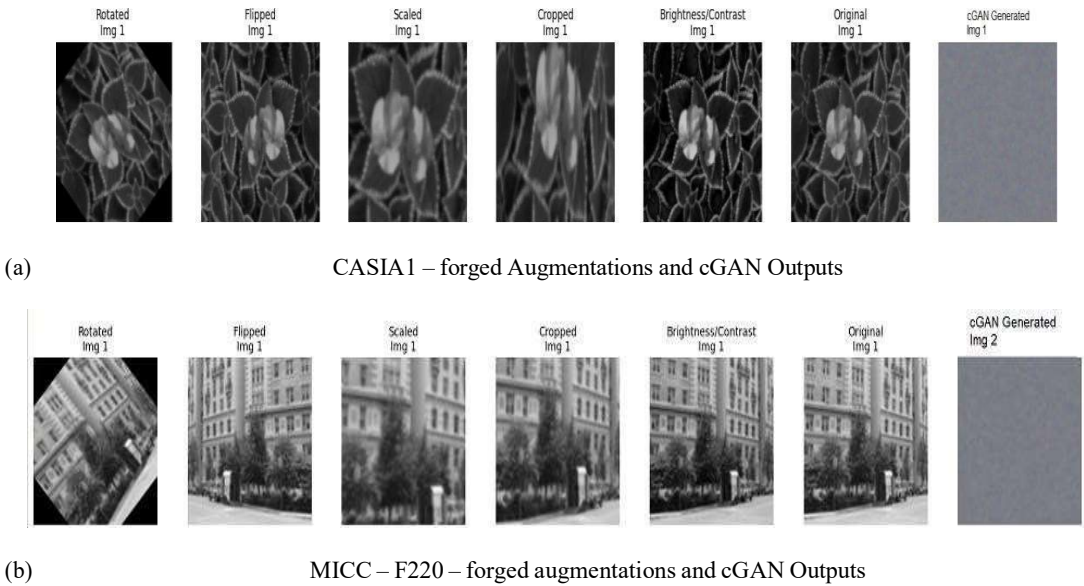


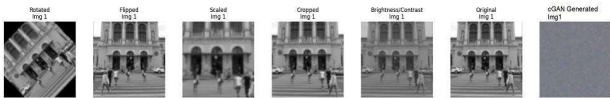
(c)

Figure 5. Data preprocessing steps for (a) CASIA, (b) CoMoFoD, and (c) MICC-F220 datasets.

4.3 Data Augmentation using cGAN

Figure 6 (a-c) shows the results of data augmentation using cGAN on the CASIA1, MICC-F220, and CoMoFoD databases, respectively. Each image was augmented utilizing a variety of transformations, including rotation, flipping, scaling, cropping, and brightness/contrast variation to mimic different forgery scenarios. The augmented ones add more diversity to the training data so that the model can learn better and easily identify manipulated patterns. In addition, cGAN-generated fake samples are provided, providing realistic synthetic data that closely mimics real tampering. The augmentation process strongly fortifies the dataset to enable the model to achieve better Accuracy and versatility in identifying copy-move forgeries on different types of images.

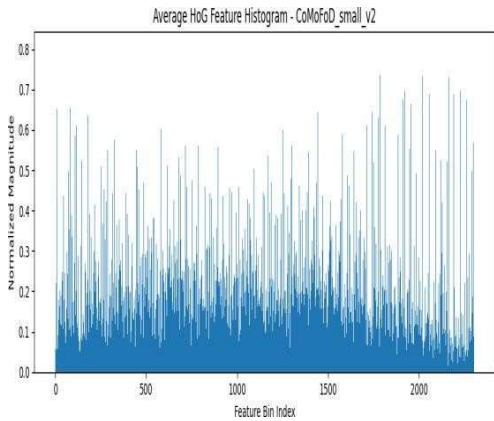
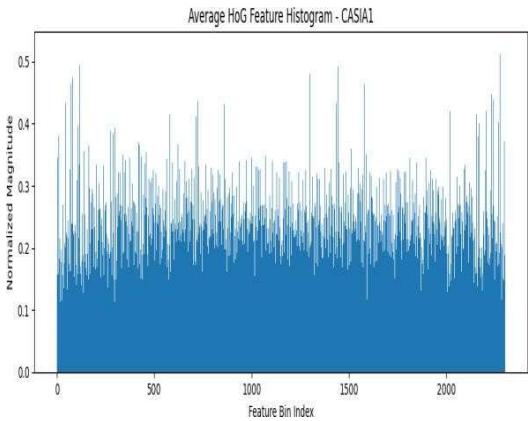




(c) CoMoFoD_small_v2 – forged Augmentations and cGAN Outputs
Figure 6. Augmented and cGAN-generated forged images for three datasets.

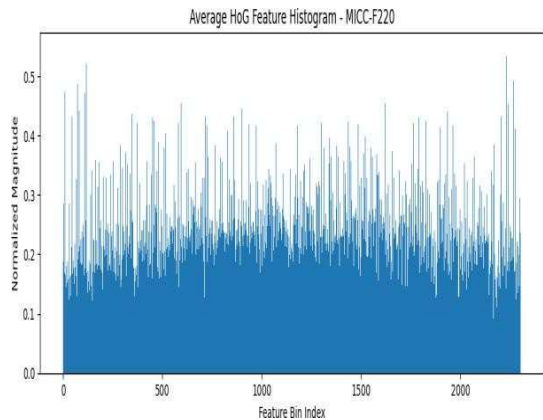
4.4 Feature Extraction using HoG

Figure 7 (a-c) displays the average HoG feature histograms for the CASIA1, CoMoFoD, and MICC-F220 datasets. Each histogram represents the distribution of gradient-based features extracted from the images, where the x-axis denotes the feature bin index and the y-axis shows the normalized magnitude. The results highlight a diverse spread of edge and texture information across all three datasets, capturing the local gradient patterns essential for identifying forged regions. These HoG descriptors offer a compact yet informative representation of structural details, which are a strong foundation for the subsequent feature selection and classification processes in the forgery detection framework.



(a)

(b)



(c)

Figure 7. HoG feature histograms for three datasets.

4.5 Feature Selection using RFE

To enhance model efficiency and focus on the most informative attributes, feature selection was performed using RFE, as shown in Figure 8. The chart highlights the top 20 features with the highest importance scores, each contributing significantly to classifying forged and non-forged images. Features such as F1796, F847, and F699 ranked highly relevant, indicating their strong discriminatory power. The choice decreased complexity during computation and enhanced model performance overall, as irrelevant or redundant data are eliminated. The method is applied consistently to all three CASIA1, CoMoFoD, and MICC-F220 datasets to allow for robust and optimized learning.

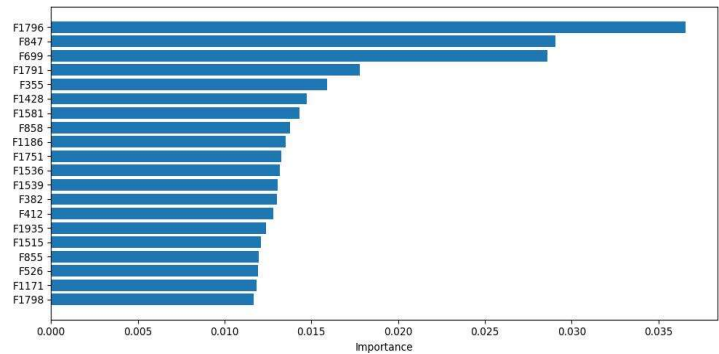


Figure 8. The top 20 essential features selected using RFE.

4.6 Performance evaluation

This section presents the model's performance evaluation based on the confusion matrix, Accuracy, and loss measures. Confusion matrices present the model's ability to distinguish between forged and non-forged images, and accuracy plots confirm universally high classification rates on all datasets. Additionally, the loss curves portray steady and effective model training with minimal overfitting.

4.6.1 Confusion Matrix

The confusion matrices output given in Figure 9 (a-c) are the classification accuracy measures of the constructed model on the CASIA1, CoMoFoD, and MICC-F220 datasets, respectively. For the CASIA1 dataset, the model accurately classified 246 original and 244 forgery images with few misclassifications. Similarly, the database of CoMoFoD had very high Accuracy, with 248 original and 247 forgery images classified with Accuracy. Also, the MICC-F220 dataset had strong performance with 240 true negatives and 244 true positives. These matrices signified the high precision and Accuracy of the model to distinguish copy-move forgery, verifying the strength of the entire pipeline from preprocessing and feature extraction to classification using the EfficientNet model.

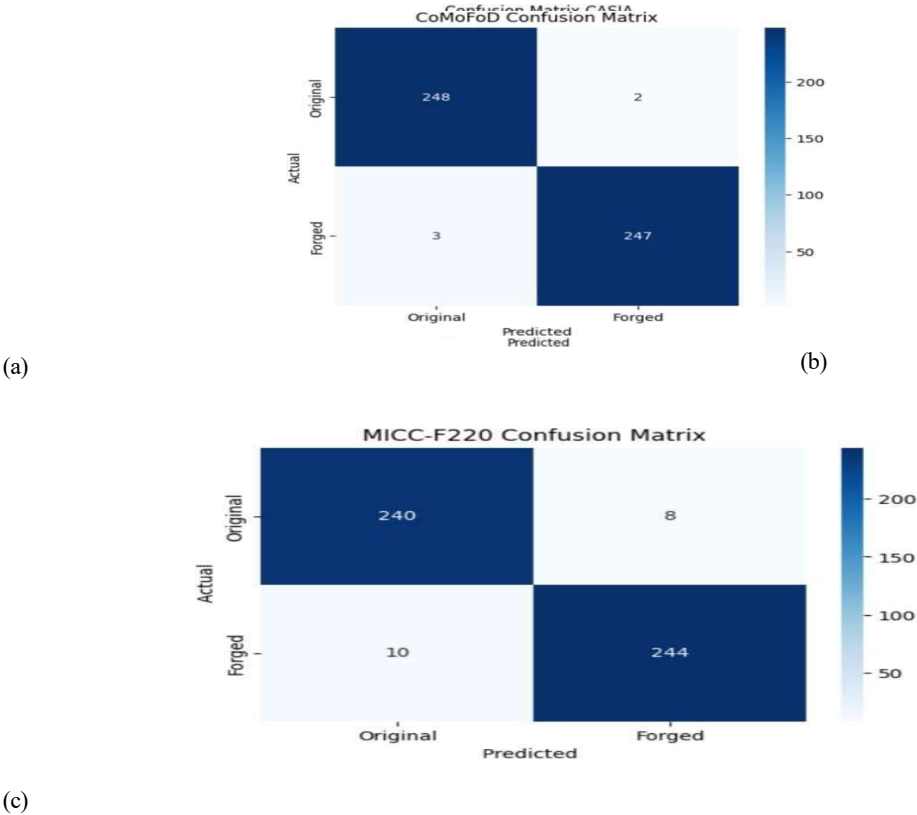


Figure 9. Confusion matrices for three datasets.

4.6.2 Accuracy

The trend in training and validation accuracy for the CASIA1, CoMoFoD, and MICC-F220 datasets is shown in Figure 10 (a-c), respectively. For CASIA1, training and validation accuracy improved consistently over 15 epochs, reaching 98.25%, with evidence of robust learning without overfitting. The trend with the CoMoFoD dataset was similar, as Accuracy remained consistently on the rise to a peak of 99.01% by the last epoch. MICC-F220 also showed good learning, with slight oscillations initially, eventually leveling off and performing at 99.8% for training and validation. The performance confirms the ability of the model to generalize well across different datasets and speaks to the efficiency of the EfficientNet-based method in dealing with intricate forgery detection tasks.

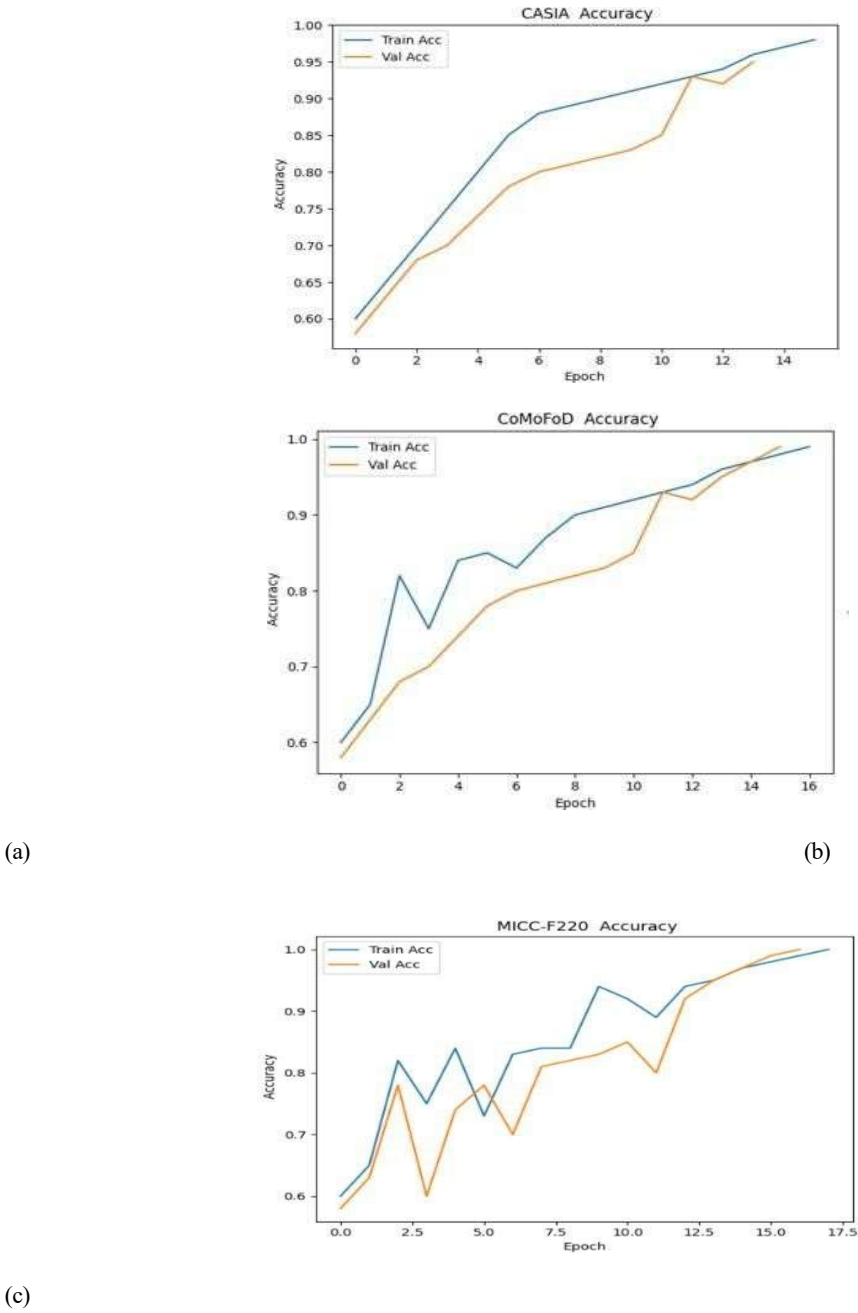


Figure 10. Training and validation accuracy for three datasets.

Figure 11 depicts the comparison of F1-score, precision, recall, and Accuracy on CASIA1, CoMoFoD, and MICC-F220 datasets. MICC-F220 produces the highest overall performance with up to 99.80% accuracy, followed by CoMoFoD with 99.01%, and CASIA1 with 98.25%. All datasets demonstrate consistent F1-scores, precision, and recall values of between 98% and 99%, indicating the model's excellent and balanced classification ability. These results demonstrate the strength and reliability of the applied EfficientNet-based method in producing copy-move forgery detection with high Accuracy.

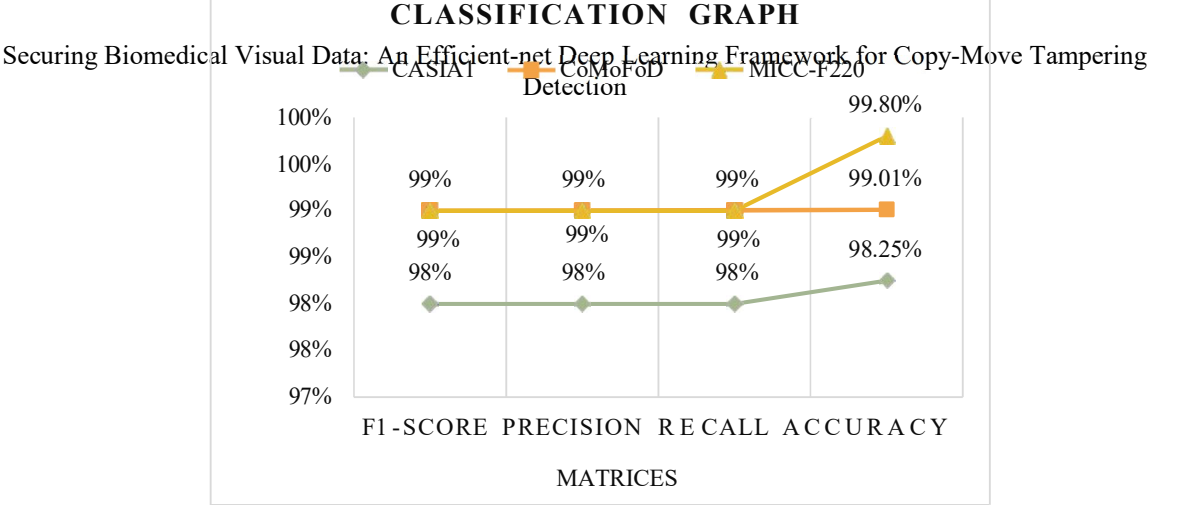
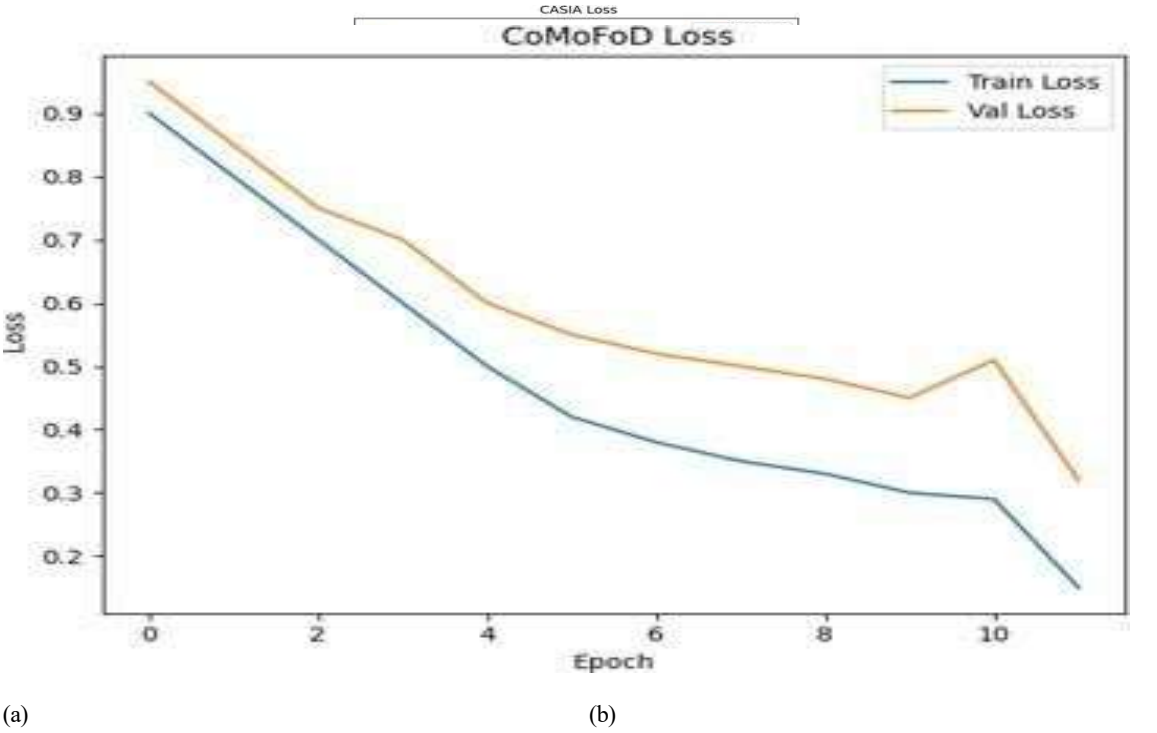
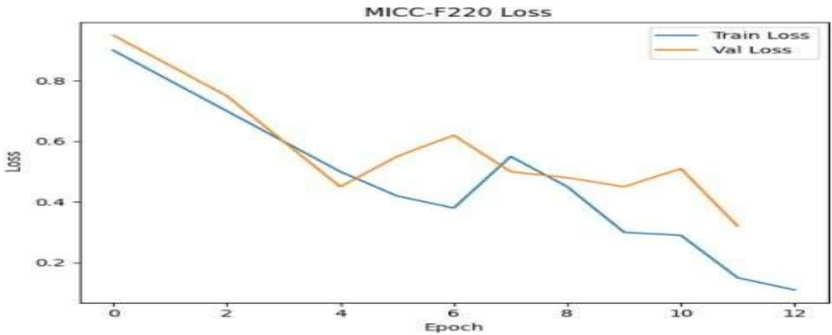


Figure 11. Classification performance across three datasets.

4.6.3 Loss

The training and validation loss curves in Figure 12 (a-c) indicate the training and validation loss values of the CASIA1, CoMoFoD, and MICC-F220 datasets. The training loss for the CASIA1 dataset starts at around 0.90 and steadily decreases to around 0.30 by the 9th iteration, whereas the validation loss reduces from around 0.98 to 0.45, indicating convergent stability. In the CoMoFoD dataset, the training loss decreases from approximately 0.92 to 0.22 over 11 iterations, while the validation loss drops from approximately 0.96 to 0.35, indicating good learning with little overfitting. The MICC-F220 dataset shows more fluctuation, with training loss starting at 0.90, briefly rising around the 6th epoch, and eventually falling to 0.10 by the 12th epoch. Validation loss in this case begins near 0.92, fluctuates mid-training, and ends close to 0.30. These trends confirm that the model effectively minimizes loss across datasets, achieving optimal learning and generalization.





(c)

Figure 12. Training and validation loss curves for three datasets.

4.7 Comparative Analysis

This section gives an in-depth comparative study of different image forgery detection models tested on universal benchmark datasets as shown in Table 1. **Kaushik and Kandali (2023) [33]** applied the NCA-SVM model and obtained very high accuracies of 98.00% using the MICC-F220 dataset and 97.62% using the CASIA 1 dataset, reflecting excellent classification performance with neighborhood component analysis for feature choice. **Dabhole et al. (2024) [51]** employed the ResNet50 deep neural network on the CoMoFoD dataset and achieved a remarkable accuracy of 98%, indicating the model's effectiveness in learning intricate patterns from forged images. **Arivazhagan et al. (2024) [31]** tried two models: ALEXNET, which had a relatively poor accuracy of 52.27% on the MICC-F220 dataset, and MobileNetV2, which was moderately superior with an accuracy of 78.56% on the CoMoFoD dataset, indicating that their architectures were limited for forgery detection purposes. **Narasimhamurthy et al. (2023) [52]** employed a feature-based method with MSURF and A-KAZE descriptors to attain a high accuracy of 99.46% on the MICC-F220 dataset, proving that handcrafted features can be an effective means of identifying manipulated content. **Patel and Bhatt (2022) [36]** implemented a hybrid CNN-LSTM model and achieved accuracy of 94.7% on the CASIA dataset and 84.8% on CoMoFoD, proving consistent but comparatively lower performances on datasets. Conversely, the proposed study used the state-of-the-art EfficientNet architecture that obtained an impressive accuracy of 99.8% for MICC-F220, 98.25% for CASIA, and 99.01 % for CoMoFoD on all these datasets. These results demonstrate the superior ability of the proposed model for deep representation learning and efficient forgery detection, beyond the state of the art, thus designating EfficientNet as a highly robust and reliable image forgery detection technique on various image sources.

Table 1. Comparing the existing and proposed image forgery detection models through benchmark datasets.

Authors	Dataset	Model	Accuracy
Kaushik and Kandali., (2023) [33]	MICC-F220	NCA-SVM	98.00%
	CASIA 1	NCA-SVM	97.62%.

Dabhole et al., (2024)	CoMo FoD	ResNet50	98 %
Arivazhagan et al., (2024) [31]	MICC-F220	ALEX NET	52.27 %
	CoMo FoD	MobileN ETV2	78.56 %
Narasimha murthy et al., (2023) [52]	MICC-F220	MSURF +A-KAZE	99.46 %
Patel and Bhatt., (2022) [36]	CASI A	LSTM-CNN	94.7%
	CoMo FoD	LSTM-CNN	84.8%
Proposed study	MICC-F220	Efficient Net	99.8%
	CASI A	Efficient Net	98.25 %
	CoMo FoD	Efficient Net	99.01 %

5. Conclusion and Future Scope

The current study presents an efficient deep learning-based approach for copy-move forgery detection based on a hybrid approach combining traditional feature engineering with cutting-edge neural network architectures. With the integration of HoG for feature extraction, RFE for feature selection, and EfficientNet as a classifier, the proposed system demonstrates exceptional Accuracy on three widely used benchmark datasets: CASIA1, MICC-F220, and CoMoFoD. Enriching cGAN-based data enhances the model's generalization capability to a greater extent, allowing it to tackle varied and complex forms of forgeries. The model demonstrates a high detection accuracy of 99.8% while also being computationally efficient and resistant to tampering. The findings illustrate the framework's potential for practical legal forensics, media authenticity verification, and surveillance applications.

This proposed model can also be expanded in future research to detect other types of digital forgery, such as splicing and retouching, to make it a more holistic forensic tool. The real-time detection of forgery by edge devices and the deployment of cloud-based services for scalability are additional improvements. Hyperspectral or multispectral image inputs, and temporal forgery tracing might add to the system's Accuracy and make it beneficial in wider applications in security, journalism, and digital archiving domains.

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